

## ORIGINAL ARTICLE OPEN ACCESS

# Assigning Phenologically Asynchronous Moths to Source Populations Using Individual Genotypes

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**Received:** 17 September 2024 | **Revised:** 20 May 2025 | **Accepted:** 2 June 2025

**Handling Editor:** Sean D Schoville

**Funding:** This work was supported by Natural Sciences and Engineering Research Council of Canada; Natural Resources Canada.

**Keywords:** dispersal | machine learning | migration | outbreaks | spatial synchrony | spruce budworm

## ABSTRACT

The spruce budworm (*Choristoneura fumiferana*; SBW) is a periodically outbreaking forest insect pest that affects the boreal forests of North America through extensive defoliation and tree mortality. Causes of widespread spatial synchrony of SBW outbreaks remain a key question in the ecology and management of this species. While the Moran effect (correlated favourable environmental conditions) and density-dependent dispersal (from epicentres of demographic explosions) have been proposed and supported as drivers of synchronised outbreaks, the relative contribution of long-distance dispersal is still poorly understood. In this study, we use a novel approach to distinguish resident from migrant moths and to assign migrants to likely source clusters with the goal of better characterising regional dispersal. First, we characterise the genetic diversity and structure of resident SBW larvae and three phenologically separated groups of moths over one flight season using Genotyping-by-Sequencing. Then, using a novel machine learning approach, we assign putative migrants to their likely source populations. We hypothesised that migrant moths and resident larvae would be genetically distinct and could be assigned to source populations. Our findings revealed complex patterns of moth dispersal and population differentiation within a single season, including two spatially overlapping genetic clusters. We observed subtle but significant genetic differences between resident larvae and migrant moths, supporting the hypothesis that long-distance dispersal contributes to outbreak dynamics and synchrony. These insights enhance our understanding of SBW population dynamics and suggest that effective management strategies, such as the Early Intervention Strategy (EIS), must account for the role of dispersal in mitigating the detrimental effects of major outbreaks.

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## 1 | Introduction

Northern forests are shaped by natural disturbances such as drought, fire, windthrow and insect outbreaks. Insect outbreaks represent one of the most significant disturbances in the Canadian boreal forest in terms of area impacted. Many defoliating pest species undergo large-scale, periodic and spatially synchronous outbreaks that negatively affect forest health and ecosystem dynamics (Boyd et al. 2013; Maclean 1984; Townsend et al. 2004), with associated economic and social consequences (Chang et al. 2012; Niquidet et al. 2016). Despite their impact and notoriety, the factors underpinning spatially complex outbreak dynamics are uncertain. More specifically, important questions remain regarding the role of dispersal in spreading and synchronising outbreaks (James et al. 2015; Larroque et al. 2019; Liebhold et al. 2004). In addition to its importance to spatial ecological theory, understanding dispersal and how it influences outbreak dynamics is fundamental to developing and implementing management strategies to mitigate outbreak impacts (Johns et al. 2019).

The eastern spruce budworm (*Choristoneura fumiferana*; hereafter SBW) (Lepidoptera: Tortricidae) is one of the most significant and persistent disturbance agents in the North American boreal forest, with evidence of outbreaks detectable back to the Holocene (Simard et al. 2006). In eastern Canada, outbreaks have a periodicity of 25–45 years and may persist in an area for between 5 and 15 years (Pureswaran et al. 2016). Damage associated with this univoltine pest is caused by the larvae, which emerge in the spring to feed (i.e., defoliate) on the developing foliage of its preferred host, balsam fir (*Abies balsamea*) and to a lesser extent spruce (*Picea* sp.). Repeated years of defoliation can cause extensive host tree death and radial growth losses, with downstream impacts on the timber industry (Chang et al. 2012). The previous regional SBW outbreak occurred from ~1967 to 1993 and caused moderate to severe damage to over 50 million hectares of forest in eastern North America (Kettela 1983). The current outbreak, which began around 2006 and has so far impacted ~15 million ha in eastern Canada, has the potential to cause a loss of as much as 43 million m<sup>3</sup> of available timber, with \$35 billion of that occurring between 2017 and 2067 (Liu et al. 2019). However, conventional foliage protection tactics (e.g., insecticide treatments) to prevent such losses are also expensive, such that even a modest level of protection of susceptible forests (e.g., 20%) could cost up to \$164 million (Liu et al. 2019). Aside from its economic impacts, conifer death caused by SBW feeding can also modify the forest landscape by creating canopy gaps and altering forest species composition (Bouchard et al. 2007; Bouchard and Pothier 2010; Kneeshaw and Bergeron 1998). Further, the spatial legacies in forest structure left behind by the spruce budworm outbreaks can result in enhanced susceptibility to other natural disturbances, such as windthrow (Taylor and MacLean 2009) and wildfire (James et al. 2017).

A key feature of SBW is its high dispersal capacity and the potential role it plays in shaping outbreak dynamics. Once larval development is completed in mid-summer, the larvae pupate and weeks later emerge to mate and lay eggs on host needles. However, prior to egg lay, the male and female moths are quite active and may undergo local or long-distance dispersal to find mates and/or new hosts Greenbank et al. (1980). In some

instances, dispersing moths may be carried via prevailing winds hundreds of km in such densities that they can be detected via weather radar Boulanger et al. (2017). Over the past 70 years, two major theories have been at the centre of debates around SBW outbreak dynamics each of which interprets the role of moth dispersal differently (for a full description see Pureswaran et al. 2016; Johns et al. 2019). In brief, the first theory, often referred to as the ‘oscillatory hypothesis’, argues that outbreak periodicity is driven by lagged density-dependent prey–predator interactions involving SBW and a broad array of generalist natural enemies. At regional levels these disparate cycles are brought into synchrony by copious moth dispersal coupled with spatially autocorrelated environmental conditions (i.e., the Moran Effect, Moran 1953; Royama 1984; Myers 1998). Importantly, under this hypothesis moth dispersal does not spread the outbreak per se but merely synchronises the oscillations of neighbouring outbreaks. The second theory, often referred to as the ‘epicentre hypothesis’ in its contemporary form, agrees with the important role of density-dependence in driving outbreak collapse, but argues for the essential role moth dispersal plays in spreading outbreaks by boosting low-density populations and thus allowing them to escape from natural constraints, such as low mating success (Régnière et al. 2012) and high predation pressure (Régnière et al. 2019; Bouchard et al. 2018). Although these hypotheses are often framed as mutually exclusive, it is likely in practice that dispersal serves a role as both outbreak synchronizer and spreader at different scales, and the challenge to researchers is to better understand under what conditions each dominates (Kendall et al. 2000; Ranta et al. 1995).

Dispersal is difficult to measure in the field, particularly for small organisms such as insects. Consequently, evidence for dispersal-induced population synchrony has largely been based on theoretical conjecture or mathematical models (Ims and Andreassen 2005). However, empirical, genetic-based methods are increasing in popularity given their ability to quantify functional connectivity among populations based on similarities in genetic diversity (Norén and Angerbjörn 2014; Schwartz et al. 2002). The genetic structure of outbreaking, geographically separate populations can be used to infer dispersal patterns, as high rates of dispersal allow for gene flow (Ibrahim et al. 1996). The use of molecular markers is particularly valuable for investigating cyclic irruptive systems, where extreme fluctuations in population density result in strong, and often repeated, genetic bottlenecks (Franklin et al. 2014). Post-outbreak isolated populations and reduced population sizes allow for genetic drift during which one expects strong genetic differentiation to emerge (James et al. 2015). Previous genetic analyses of known resident populations have confirmed the importance of dispersal in the spatial synchrony of SBW outbreaks (James et al. 2015; Larroque et al. 2020, 2019). However, we know less about the within-season spatial dynamics of outbreaks (including dispersal) and the extent to which distinguishing resident individuals from moths that may have arrived at location via dispersal (i.e., putative migrants) can help us to better characterise synchronous outbreak dynamics.

In this study, we present a novel, phenologically based approach for distinguishing resident from migrant moths and for the application of population genetics to these differentiated

populations to better understand the role of dispersal in SBW outbreaks. Specifically, we use genotypic data for SBW larvae and moths collected at three different times within a single season to characterise the role of long-distance dispersal in outbreak spatial synchrony. Whereas previous research using resident larval SBW (i.e., Larroque et al. 2019) provided insight into connectivity among local populations, the complementary moth samples investigated here provide information on moth residents as well as putative early and late migrants. We expect these groupings to be active at different times due to regional variations in climate and associated flight phenology. Our goals were to: (1) characterise and compare spatial genetic structure of SBW larvae and three phenologically separated groups of moths within a single season; and (2) apply a novel machine learning assignment approach to distinguish residents from migrants and to assign putative immigrants to their source population. We hypothesise that phenologically classified putative migrant moths will be distinguishable from local larval residents based on genotypes. We also expect these differences to be subtle given the high population densities and outbreak status of the region when sampling was undertaken, because high gene flow can reduce population differentiation. Through the process of assigning putative migrants to likely source locations or regions, we can develop a better understanding of the relative importance of local population processes and dispersal in driving synchronous outbreaks.

## 2 | Methods

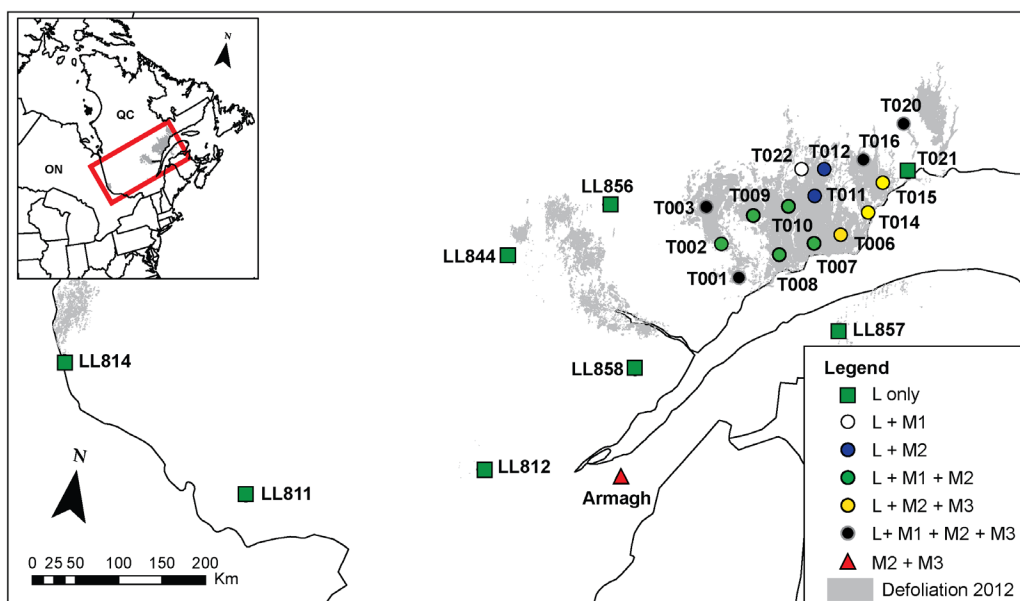
### 2.1 | Study Area and Sample Collection

The outbreak examined in this current study was first recorded in 2006 and is located along the northern shore of the St. Lawrence River and the Quebec-Ontario border in Canada. Insects were collected over 32 sites within this area (Figure 1).

Spruce budworm larvae and adult male moths were collected in the summer of 2012. Mature larvae (4th–6th instar; L) were collected in early to mid-June in the Côte Nord region (Figure 1) and raised to maturity on synthetic McMorran diet in the lab (McMorran 1965) using standard rearing protocols (Ebling and Dedes 2014). Larvae collected elsewhere were reared to moth in mesh bags on live foliage. Adult moths were captured using pheromone-baited Uni-Traps containing an insecticidal strip. Traps were emptied at three different times over the course of the season with the goal of capturing early migrant moths, local resident moths and late migrant moths.

Trap deployment and collection was guided by expected phenological timing as predicted using BioSIM-10 (Régnière et al. 2014). BioSIM uses available weather station data and accounts for station proximity, topography and local thermal gradients to generate daily maps of minimum and maximum temperatures (Régnière 1996; Regniere and Bolstad 1994). In our case, these temperature fields were then used to drive a previously applied (Gandiaga and James 2023; Garcia et al. 2022), and validated (Candau et al. 2018) spruce budworm phenology model (Régnière 1996; Régnière et al. 2012; Régnière and You 1991).

Pheromone trapping is a common method used to monitor adult male abundance over large areas. To obtain the SBW moths for our study, we used samples collected from pheromone-baited moth traps deployed throughout the area of outbreak in Quebec, Canada (Figure 1). Typically, these traps are deployed prior to moth emergence and collected once at the end of the expected flight period to obtain a total moth count for the area. However, because trap samples are collected at the end of the flight season, the total moth count can potentially include both resident and immigrant moths. Here, we used multiple collections throughout the flight season and genomic analyses to improve our ability to distinguish resident from migrant moths and thus clarify the potential role of dispersal in outbreak dynamics. Traps were



**FIGURE 1** | Study area indicating sample locations and extent of defoliation at the time of data collection. Upper left inset indicates study area (red box) in a North American context. Details for each sample location are found in Table 1.

**TABLE 1** | Details of each site sampled in 2012, including geographic coordinates, number of individuals genotyped for each site—form combination (*N*), observed heterozygosity (*Ho*), expected heterozygosity (*He*) and mean allelic richness (*Ar*).

| ID | Sample name | <i>N</i> | Site   | Form | <i>Ho</i> | <i>He</i> | <i>Ar</i> | Lat.  | Long.  | Region           |
|----|-------------|----------|--------|------|-----------|-----------|-----------|-------|--------|------------------|
| 1  | Armagh_M2   | 19       | Armagh | M2   | 0.17      | 0.20      | 1.20      | 46.75 | −70.58 | CentralQC        |
| 2  | Armagh_M3   | 23       | Armagh | M3   | 0.18      | 0.21      | 1.21      | 46.75 | −70.58 | CentralQC        |
| 3  | LL811_L     | 25       | LL811  | L    | 0.16      | 0.19      | 1.19      | 45.89 | −76.03 | WesternQC        |
| 4  | LL812_L     | 24       | LL812  | L    | 0.17      | 0.21      | 1.21      | 46.62 | −72.64 | CentralQC        |
| 5  | LL814_L     | 16       | LL814  | L    | 0.17      | 0.21      | 1.21      | 46.77 | −79.12 | WesternQC        |
| 6  | LL844_L     | 21       | LL844  | L    | 0.16      | 0.19      | 1.19      | 48.84 | −72.83 | CentralQC        |
| 7  | LL856_L     | 8        | LL856  | L    | 0.17      | 0.20      | 1.20      | 49.53 | −71.37 | CentralQC        |
| 8  | LL857_L     | 17       | LL857  | L    | 0.17      | 0.20      | 1.20      | 48.51 | −67.56 | Gaspe            |
| 9  | LL858_L     | 12       | LL858  | L    | 0.16      | 0.19      | 1.19      | 47.88 | −70.61 | CentralQC        |
| 10 | T001_M1     | 5        | T001   | M1   | 0.19      | 0.23      | 1.22      | 48.96 | −69.19 | Lower_CoteNord   |
| 11 | T001_M2     | 15       | T001   | M2   | 0.20      | 0.24      | 1.24      | 48.96 | −69.19 | Lower_CoteNord   |
| 12 | T002_L      | 23       | T002*  | L    | 0.21      | 0.24      | 1.24      | 49.28 | −69.53 | Lower_CoteNord   |
| 13 | T002_M1     | 17       | T002   | M1   | 0.19      | 0.22      | 1.22      | 49.28 | −69.53 | Lower_CoteNord   |
| 14 | T002_M2     | 15       | T002   | M2   | 0.19      | 0.22      | 1.21      | 49.28 | −69.53 | Lower_CoteNord   |
| 15 | T003_L      | 19       | T003*  | L    | 0.21      | 0.25      | 1.25      | 49.65 | −69.84 | Lower_CoteNord   |
| 16 | T003_M1     | 2        | T003   | M1   | 0.20      | 0.25      | 1.23      | 49.65 | −69.84 | Lower_CoteNord   |
| 17 | T003_M2     | 14       | T003   | M2   | 0.18      | 0.21      | 1.21      | 49.65 | −69.84 | Lower_CoteNord   |
| 18 | T003_M3     | 1        | T003   | M3   | 0.21      | NA        | 1.21      | 49.65 | −69.84 | Lower_CoteNord   |
| 19 | T006_L      | 25       | T006*  | L    | 0.20      | 0.24      | 1.24      | 49.51 | −67.66 | Central_CoteNord |
| 20 | T006_M2     | 16       | T006   | M2   | 0.19      | 0.23      | 1.22      | 49.51 | −67.66 | Central_CoteNord |
| 21 | T006_M3     | 1        | T006   | M3   | 0.20      | NA        | 1.20      | 49.51 | −67.66 | Central_CoteNord |
| 22 | T007_L      | 10       | T007   | L    | 0.17      | 0.21      | 1.21      | 49.39 | −68.07 | Central_CoteNord |
| 23 | T007_M1     | 1        | T007   | M1   | 0.22      | NA        | 1.22      | 49.39 | −68.07 | Central_CoteNord |
| 24 | T007_M2     | 14       | T007   | M2   | 0.19      | 0.23      | 1.23      | 49.39 | −68.07 | Central_CoteNord |
| 25 | T008_L      | 24       | T008*  | L    | 0.19      | 0.24      | 1.23      | 49.24 | −68.60 | Lower_CoteNord   |
| 26 | T008_M1     | 2        | T008   | M1   | 0.15      | 0.18      | 1.17      | 49.24 | −68.60 | Lower_CoteNord   |
| 27 | T008_M2     | 9        | T008   | M2   | 0.18      | 0.22      | 1.22      | 49.24 | −68.60 | Lower_CoteNord   |
| 28 | T009_L      | 24       | T009*  | L    | 0.20      | 0.24      | 1.24      | 49.61 | −69.08 | Lower_CoteNord   |
| 29 | T009_M1     | 6        | T009   | M1   | 0.18      | 0.22      | 1.21      | 49.61 | −69.08 | Lower_CoteNord   |
| 30 | T009_M2     | 12       | T009   | M2   | 0.19      | 0.23      | 1.23      | 49.61 | −69.08 | Lower_CoteNord   |
| 31 | T010_L      | 24       | T010*  | L    | 0.21      | 0.24      | 1.24      | 49.75 | −68.54 | Central_CoteNord |
| 32 | T010_M1     | 14       | T010   | M1   | 0.20      | 0.23      | 1.23      | 49.75 | −68.54 | Central_CoteNord |
| 33 | T010_M2     | 16       | T010   | M2   | 0.21      | 0.23      | 1.23      | 49.75 | −68.54 | Central_CoteNord |
| 34 | T011_L      | 10       | T011   | L    | 0.16      | 0.20      | 1.20      | 49.89 | −68.13 | Central_CoteNord |
| 35 | T011_M2     | 15       | T011   | M2   | 0.20      | 0.24      | 1.24      | 49.89 | −68.13 | Central_CoteNord |
| 36 | T012_L      | 15       | T012*  | L    | 0.17      | 0.21      | 1.21      | 50.17 | −68.02 | Central_CoteNord |
| 37 | T012_M2     | 10       | T012   | M2   | 0.20      | 0.23      | 1.23      | 50.17 | −68.02 | Central_CoteNord |
| 38 | T014_L      | 25       | T014*  | L    | 0.19      | 0.23      | 1.23      | 49.77 | −67.25 | Upper_CoteNord   |

(Continues)

TABLE 1 | (Continued)

| ID | Sample name | N  | Site  | Form | Ho   | He   | Ar   | Lat.  | Long.  | Region           |
|----|-------------|----|-------|------|------|------|------|-------|--------|------------------|
| 39 | T014_M2     | 16 | T014  | M2   | 0.18 | 0.22 | 1.22 | 49.77 | −67.25 | Upper_CoteNord   |
| 40 | T014_M3     | 1  | T014  | M3   | 0.16 | NA   | 1.16 | 49.77 | −67.25 | Upper_CoteNord   |
| 41 | T015_L      | 4  | T015  | L    | 0.15 | 0.18 | 1.17 | 50.09 | −67.05 | Upper_CoteNord   |
| 42 | T015_M2     | 18 | T015  | M2   | 0.18 | 0.23 | 1.22 | 50.09 | −67.05 | Upper_CoteNord   |
| 43 | T015_M3     | 5  | T015  | M3   | 0.24 | 0.24 | 1.24 | 50.09 | −67.05 | Upper_CoteNord   |
| 44 | T016_L      | 25 | T016* | L    | 0.20 | 0.23 | 1.23 | 50.31 | −67.40 | Upper_CoteNord   |
| 45 | T016_M1     | 3  | T016  | M1   | 0.23 | 0.23 | 1.23 | 50.31 | −67.40 | Upper_CoteNord   |
| 46 | T016_M2     | 16 | T016  | M2   | 0.23 | 0.25 | 1.25 | 50.31 | −67.40 | Upper_CoteNord   |
| 47 | T016_M3     | 2  | T016  | M3   | 0.28 | 0.25 | 1.26 | 50.31 | −67.40 | Upper_CoteNord   |
| 48 | T020_L      | 7  | T020  | L    | 0.17 | 0.20 | 1.20 | 50.72 | −66.80 | Upper_CoteNord   |
| 49 | T020_M1     | 1  | T020  | M1   | 0.18 | NA   | 1.18 | 50.72 | −66.80 | Upper_CoteNord   |
| 50 | T021_M2     | 18 | T021  | M2   | 0.20 | 0.24 | 1.24 | 50.22 | −66.66 | Upper_CoteNord   |
| 51 | T022_L      | 9  | T022  | L    | 0.17 | 0.19 | 1.19 | 50.15 | −68.39 | Central_CoteNord |
| 52 | T022_M1     | 16 | T022  | M1   | 0.17 | 0.21 | 1.21 | 50.15 | −68.39 | Central_CoteNord |
| 53 | T001_L      | 43 | T001* | L    | 0.18 | 0.21 | 1.21 | 48.94 | −69.18 | Lower_CoteNord   |
| 54 | T021_L      | 33 | T021* | L    | 0.17 | 0.20 | 1.20 | 50.19 | −66.66 | Upper_CoteNord   |
| 55 | T020_M2     | 38 | T020  | M2   | 0.21 | 0.25 | 1.25 | 50.78 | −66.97 | Upper_CoteNord   |
| 56 | T020_M3     | 1  | T020  | M3   | 0.22 | NA   | 1.22 | 50.72 | −66.80 | Upper_CoteNord   |

*Note:* *Form* refers to both life-stage (larvae [L] vs. moth [M]) and phenological timing of moth collection: M1 refers to early putative migrant moths, M2 refers to locally phenologically synchronous moths, and M3 refers to late putative migrant moths. See text for details. Sites are shown in a geographic context in Figure 1. Sites with < 5 individuals were not used in our population-level analyses. All individuals from M1 and M3 sites were used for assignment, regardless of the number of individuals collected. Sites marked with an asterisk (\*) include larvae that were differentiated by sex (Figure 4).

initially deployed in early June, ~2 weeks before the expected moth flight period, and emptied three times during the summer. The latest collection was mid-August. Collection timing occurred up to 2 weeks following trap installation with the goal of capturing the coarse categories of migrants and residents. Given the logistical challenges of visiting all of these traps multiple times over such a broad area, the date on which we collected the specimens from the trap may not reflect the date on which the moths entered the trap. However, we are confident that we were able to capture the coarse scale differences among our three phenological categories. We first targeted early migrant moths (M1) by emptying the traps 1–2 weeks prior to the phenologically predicted period (99% confidence interval) of moth emergence. Traps were emptied a second time within a week of the end of the period of predicted moth emergence (M2). Finally, moths (if present) were collected from traps a third time (M3) approximately 2 weeks after M2 collection. Due to logistic constraints, including significant distances between sites, shifting phenology and unexpected site disturbances (e.g., forest fires) it was not possible to visit all sites for all three sampling periods. Pheromone trapped moths from the Armagh site (Table 1) were generously provided by Dr. Jacques Régnière (Canadian Forest Service, NRCan).

Final classification of captured male moths as M1, M2 or M3 was determined using observed (vs. predicted) weather. Using BioSim, we calculated the 99% confidence interval of expected

emergence timing for each sample site. This final classification step also used information about the timing of trap deployment and the timing of subsequent trap emptying to classify moths as residents or migrants. Moths that were presumed to have entered the trap before the 0.5th percentile of predicted local moth emergence were classified as early putative migrants (M1). M1 moths are individuals that were phenologically ahead of the local conditions. Moths that entered the trap during the period of the 99% confidence interval were classified as residents that had emerged from local larvae (M2). M2 moths are those we considered phenologically in sync with local conditions. Finally, moths that entered the trap after the 99.5 percentile of predicted local moth emergence were deemed to be putative late migrants (M3). M3 moths are individuals that were phenologically behind local conditions. Further details of the sampling procedure and timing of trap collection can be found in Gandiaga and James (2023).

## 2.2 | DNA Extraction

DNA extraction, sample preparation, sequencing and bioinformatic processing follow the methods described by Larroque et al. (2019). All larvae were reared to adults in the laboratory prior to DNA extraction. Emerging moths were placed into 1.5-mL tubes and stored at −80°C. Moth wings and abdomens were removed via dissection during which forceps and scalpels

were flame sterilised (Bunsen burner) between individuals to prevent cross-contamination. A subset of individuals was sexed during dissection. DNA was extracted from all individuals using Qiagen DNA Blood and Tissue kits (Venlo, Limburg, NL).

## 2.3 | Sequencing

DNA was prepared for Genotyping-By-Sequencing (GBS; Elshire et al. 2011) using methods described in Brunet et al. (2017). PstI-MspI GBS libraires (96-plex) were prepared by the Institut de Biologie Intégrative et des Systèmes (IBIS) at Université Laval (Quebec City, QC) using the protocol of Poland et al. (2012). Following final amplification of the pooled, adapter-ligated restriction fragments, 600 ng of each amplified library was normalised to remove the repetitive fraction by treatment with duplex-specific nuclease (Zhulidov et al. 2004). An additional PCR step using a selective reverse primer extending a single base (C) into the insert past the 3' restriction site was used to selectively amplify one-quarter of the total number of fragments, thereby increasing the read depth of sequenced fragments (Sonah et al. 2013). Single-end sequencing (100 bp reads) of these libraries was then performed with an Illumina HiSeq2000 (McGill University- Génome Québec Innovation Centre, Montreal, QC).

We used the Fast-GBS pipeline (Torkamaneh et al. 2017) for bioinformatic processing of reads. The Fast-GBS pipeline includes demultiplexing, trimming, mapping and variant calling steps to process GBS samples and provides high genotyping accuracy relative to the other pipelines (Torkamaneh et al. 2016). After demultiplexing and adapter trimming, reads smaller than 50 bp were discarded. Remaining reads were aligned to the spruce budworm reference genome (bw6 version; Larroque et al. 2020) using the Burrows–Wheeler Aligner (Li and Durbin 2010) with the *mem* algorithm. SNP calls were made as part of the Fast-GBS pipeline using Platypus (v0.8.1; Rimmer et al. 2014). Minimum read depth was set to eight. Only bi-allelic SNPs with fewer than 50% missing genotypes among all samples were retained. Finally, as some analytical methods that we use cannot handle missing data, missing genotypes were imputed using the software BEAGLE (v3.3.2; Browning and Browning 2016). This technique replaces missing genotypes with the most frequently observed genotype associated with proximal SNP loci. Variants with  $R^2$  (i.e., imputation accuracy)  $< 0.4$  were removed (Browning and Browning 2016).

## 2.4 | Filtering SNP Loci

SNP loci were filtered to minimise sequencing and genotyping errors. SNPs were first excluded based on allele frequency such that only SNPs with a minor allele frequency (MAF)  $> 5\%$  were retained. Highly correlated alleles were then identified using the *snpdgsLDpruning* function of *SNPRelate* in R (Zheng et al. 2012). Those found in high linkage disequilibrium, defined by  $R^2 > 0.2$ , were discarded due to redundancy. Filtering for MAF reduced the SNP dataset from 3246 to 2553 loci. Filtering out SNPs in high linkage disequilibrium further reduced the dataset to 2264 loci. We did not filter based on Hardy–Weinberg equilibrium (HWE) or potential selection as

loci exhibiting such non-equilibrium patterns in this spatially dynamic system are potentially useful in identifying migrant sources.

## 2.5 | Genetic Diversity

The genetic diversity measures of allelic richness, expected heterozygosity and observed heterozygosity were calculated using the *hierfstat* R package (Goudet 2005). Measures of population genetic differentiation ( $F$ -statistics) were estimated using *Fstat* (Goudet 2001) for five different groupings: (1) all samples; (2) larvae (L); (3) early moths (M1); (4) resident moths (M2); and (5) late moths (M3).

To visualise variation among groups, a principal components analysis (PCA) was performed on a matrix of Hellinger transformed (Legendre and Gallagher 2001) population-level allele frequencies using the *rda* function in the *vegan* package in R. Populations with fewer than five individuals were excluded from this and all further analyses, with the exception of the assignment analysis which specifically investigated the origin of individual early and late sampled moths. Significance of resulting PC axes was assessed using the *broken stick* criterion (Jackson 1993). We also examined differences between male and female moths reared from larvae using a subset of individuals that had been sexed during dissection (73 male, 77 female; 11 sites). The influence of life stage and site (Table 1) on population-level genetic variation was assessed using permutational ANOVA implemented in the *adonis2* function of the *vegan* package in R. This implementation of the permutational ANOVA is equivalent to a distance-based redundancy analysis (Legendre and Anderson 1999). Here, the response variable was a matrix of pairwise  $F_{st}$  values (Weir and Cockerham 1984) as implemented in the *pairwise.WCfst* function in the *hierfstat* package in R. Prior to analysis, the  $F_{st}$  distance matrix was square root transformed to ensure that distances were Euclidean (James et al. 2011).

Within our sampling strategy, we assumed that reared larvae and resident moths (M2) represent the same population. To verify this assumption, we assessed concordance in genetic structure between the L and M2 life stages using the subset of sites where both were present using a Procrustes rotation test (Peres-Neto and Jackson 2001), as implemented in the *protest* function in the *vegan* package in R. As above, data matrices were Hellinger transformed prior to ordination. Due to the relative paucity of individuals in the M1 and M3 groups, as well as their limited geographic representation (i.e., putative migrants were captured at few sites; Table 1), we did not compare these two life stages.

## 2.6 | Clustering

Genetic structure was evaluated using Discriminant Analysis of Principal Components (DAPC) implemented in the *adegenet* R package (Jombart 2008). This approach maximises between-group variance while minimising within-group genetic variance (Jombart et al. 2010). Groups (or clusters) are not defined a priori. Ward's hierarchical clustering was used

to identify the optimal number of clusters through minimisation of the Bayesian Information Criterion (BIC). To optimise the number of principal components (PC) used in the DAPC, we used cross-validation performed with the *xvalDapc* function. This approach prevents overfitting of discriminant functions while retaining enough to maximise explanatory power. The DAPC was conducted using the optimal number of clusters and PCs. This process was applied to all samples as well as the four sub-groupings described above (i.e., L, M1, M2, M3).

## 2.7 | Population Assignment

Assignment of putative migrants (M1 and M3 moths) to their most likely source clusters was performed using the *assignPOP* package in R (Chen et al. 2018). *assignPOP* uses genetic data in combination with supervised machine learning methods to predict the most likely source of unknown individuals. We also assigned putative resident M2 moths to assess the reliability of our phenological classifications. We would expect M2 moths to be assigned to clusters similarly to larvae, whereas we would expect the putative migrants to be assigned differently. This analysis is complementary to our comparison of L and M2 stages using the Procrustes rotation test described above.

This approach to population assignment is distinct from clustering of moths and larvae using DAPC. Clustering using DAPC (or other techniques) is a form of *unsupervised* classification (James et al. 2013) in which the goal is to group similar individuals. In contrast, *assignPOP* is used to train a model based on known residents (i.e., larvae) that represent the ‘true’ genetic clusters. This model is then used to assign unknown individuals to resident clusters in a form of *supervised* classification.

The procedure began with the larval resident genetic clusters identified using DAPC ( $n=2$ ). In total, 471 larvae ( $n=314$  in **Cluster A**;  $n=157$  in **Cluster B**) were used to fit this model, using 2264 loci. Next, larval genotype data and these known clusters were used to train and validate a linear discriminant analysis (*lda*; James et al. 2013) classification model within *assignPOP*. This model was then used to assign M1, M2 and M3 individuals to the larval clusters. A Chi-square test was used to determine if the assignment of moths to the different resident

larval clusters varied between life stages. We also used Fisher’s Exact Test to determine if larvae and M2 moths belonged to the same clusters.

## 3 | Results

### 3.1 | Genetic Diversity

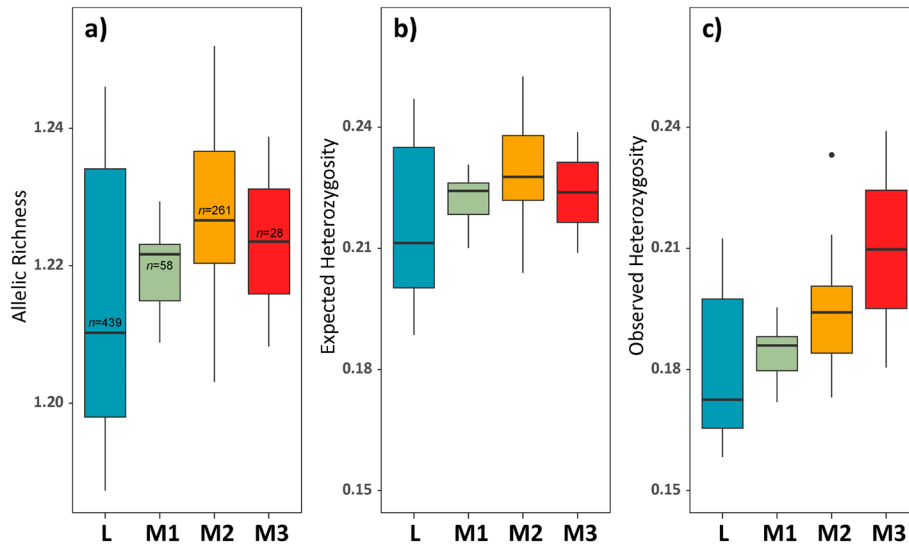
In total, we analysed genetic variation in 828 individual spruce budworm (Table 2). Genetic diversity varied somewhat among life stages and phenological groupings (Figure 2). Larvae exhibited the lowest site-level median allelic richness (**Ar**), but also the greatest variation among sites (Table 1). Larvae remained the most variable for both heterozygosity categories. Expected heterozygosity (**He**) varied little among groupings, whereas observed heterozygosity (**Ho**) was highest for the M3 category. Using ANOVA, we did not identify any significant differences among budworm groupings in terms of **Ar** ( $F_{3,41}=2.206$ ,  $p=0.102$ ) or **He** ( $F_{3,41}=2.264$ ,  $p=0.095$ ). We did find evidence for differences in **Ho** among groups ( $F_{3,41}=3.457$ ,  $p=0.025$ ). Although the M3 category seems to be unique among life stages (Figure 2), a post hoc test (Tukey’s HSD) was not able to identify any significant pair-wise comparisons despite the globally significant model. Because M3 moths were only present in two sites, and the overall analysis was unbalanced, we reran this analysis without the M3 individuals. Here, we identified a significant effect of budworm grouping on **Ar** ( $F_{2,40}=3.318$ ,  $p=0.046$ ), **He** ( $F_{2,40}=3.428$ ,  $p=0.042$ ) and **Ho** ( $F_{2,40}=3.886$ ,  $p=0.029$ ). In all cases, the comparison driving this significant effect was the difference between the L and M2 groupings (Tukey’s HSD,  $p<0.05$ ).

The differences between L and M2 groupings were unexpected. To further explore this difference, we examined differences between males and female larvae for a subset of individuals for which we had this information (Table 1). Each of the 11 sites used had seven or eight individuals of each sex. We tested the hypothesis that a difference in genetic diversity between the sexes could help us to understand the differences in diversity between M2 (males only) and L (male and female) categories described above. Using ANOVA, we found no evidence of difference in genetic diversity measures between male and female larvae (**Ar**:  $F_{1,17}=0.10$ ,  $p=0.75$ ; **He**:  $F_{1,17}=0.34$ ,  $p=0.56$ ; **Ho**:  $F_{1,17}=1.41$ ,  $p=0.25$ ).

**TABLE 2** | Genetic diversity and spatial genetic structure (*Fst*) by spruce budworm life stage.

| Stage     | <i>n</i> | <i>Ho</i>     | <i>He</i>     | <i>Ar</i>     | <i>Fst</i> | <i>p</i> |
|-----------|----------|---------------|---------------|---------------|------------|----------|
| All       | 828      | 0.187 ± 0.019 | 0.221 ± 0.017 | 1.200 ± 0.017 | 0.011      | 0.01     |
| Larvae    | 464      | 0.179 ± 0.020 | 0.215 ± 0.021 | 1.213 ± 0.021 | 0.006      | 0.01     |
| All moths | 364      | 0.194 ± 0.017 | 0.227 ± 0.012 | 1.226 ± 0.013 | 0.007      | 0.01     |
| M1        | 58       | 0.184 ± 0.009 | 0.222 ± 0.008 | 1.220 ± 0.008 | 0.0001     | 0.19     |
| M2        | 278      | 0.195 ± 0.015 | 0.229 ± 0.013 | 1.228 ± 0.013 | 0.005      | 0.01     |
| M3        | 28       | 0.210 ± 0.041 | 0.224 ± 0.021 | 1.224 ± 0.022 | 0.018      | 0.01     |

Note: *Fst* was calculated using sites indicated in Table 1. Values (±SD) of observed heterozygosity (*Ho*), expected heterozygosity (*He*) and allelic richness (*Ar*) for each population are averaged over each stage and were calculated using sample sites with greater than five individuals (Table 1). Final *p*-value indicates significance of calculated *Fst*.



**FIGURE 2** | Genetic diversity of larvae and different phenological classification of moths, including (a) allelic richness, (b) expected heterozygosity, and (c) observed heterozygosity. All diversity measures were calculated using data from locations with more than five individuals (Table 1). Number of individuals used for each category are indicated in panel 1; the same numbers apply to panels (b and c). None of the inter-group comparisons within each diversity category was significant ( $p > 0.05$ ).

**TABLE 3** | Summary of the influence of the categorical predictor *Region* (Table 1) on spatial patterns of genetic diversity, including allelic richness (Ar), expected heterozygosity (He) and observed heterozygosity (Ho).

| Form | n   | Ar   |                      | He    |                      | Ho   |                    |
|------|-----|------|----------------------|-------|----------------------|------|--------------------|
|      |     | F    | p                    | F     | p                    | F    | p                  |
| All  | 828 | 3.85 | 0.007 <sup>a,b</sup> | 3.97  | 0.006 <sup>a,b</sup> | 3.29 | 0.016 <sup>b</sup> |
| L    | 464 | 2.86 | 0.049                | 2.882 | 0.048 <sup>b</sup>   | 2.84 | 0.050 <sup>b</sup> |
| M2   | 364 | 4.32 | 0.028 <sup>a</sup>   | 4.35  | 0.027 <sup>a</sup>   | 1.54 | 0.255              |

Note: F-statistics and associated p-values are from the different ANOVA models of diversity as a function of *Region*. We only show results for all samples, larvae (L) and 'resident' moths (M2). Tukey's post hoc test was used to determine pair-wise comparisons when the global ANOVA model was significant.

<sup>a</sup>Post hoc difference identified between Upper-Cote Nord and Central-QC ( $p < 0.05$ ).

<sup>b</sup>Post hoc difference identified between Lower-Côte Nord and Central-QC ( $p < 0.05$ ).

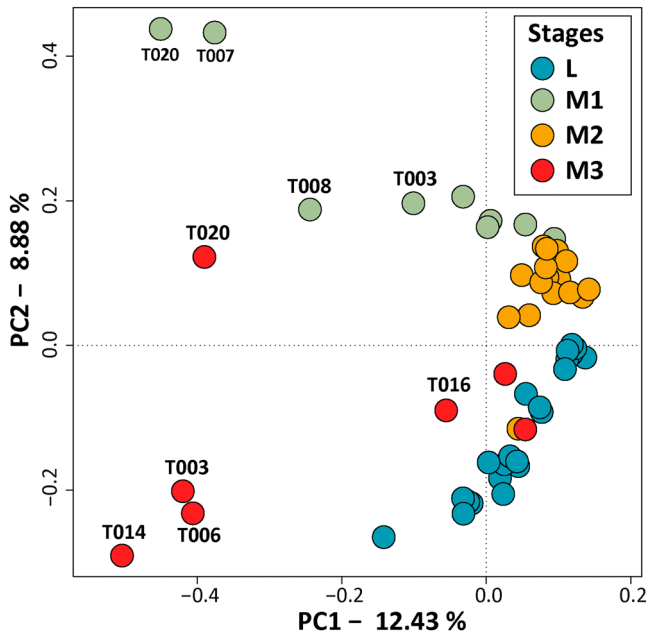
Geography as captured by *Region* (Table 1) also influenced genetic diversity for L and M2 groupings (Table 3). M1 and M3 moths were not considered here due to small sample size. We found evidence that *Region* influences allelic richness when examining L and M2 categories together and M2 moths alone, but not when examining L alone. Expected and observed heterozygosity exhibited similar patterns with marginally ( $p \approx 0.05$ ) significant differences among larvae, significant differences when examining M2 moths alone, and the combined larvae and M2 moths. Using Tukey's HSD post hoc test, we identified significant pair-wise differences between Lower-Cote Nord and Central-Quebec samples in larvae and between Upper-Cote Nord and Central-Quebec samples in M2 (Table 3).

### 3.2 | Genetic Structure

Global *Fst* based on geographic populations (*Sites*) was consistently low, but significant for all budworm groupings, except for early migrant (M1) moths (Table 2). Using principal components analysis (Figure 3) we identified differences among groupings. This visual clustering among life stages was supported by a

permutational ANOVA on a square root transformed pairwise matrix of *Fst* values ( $F_{3,41} = 19.88$ ,  $p = 0.001$ ,  $R^2 = 0.59$ ). Pairwise *Fst* values for all resident larvae showed additional subtle differences among sites, and that the strongest pairwise differences were found between sites LL856 and T020 and T010, and between T006 and LL811 (Figure S1). Similar to our analysis of genetic diversity, we also found an effect of *Region* on differentiation in larvae ( $F_{5,16} = 2.12$ ,  $p = 0.04$ ,  $R^2 = 0.39$ ) and in M2 moths ( $F_{3,12} = 6.03$ ,  $p = 0.008$ ,  $R^2 = 0.60$ ), although these effects were not as pronounced as that of grouping. As above, M1 and M3 moths were excluded here due to the small number of sites with enough samples.

We directly evaluated the correlation between L and M2 stages using a Procrustes rotation test. In doing so, we test the assumption that larvae and M2 moths represent the same resident population. Contrary to expectation, we identified a weak and non-significant correlation between L and M2 samples ( $r = 0.364$ ,  $p = 0.239$ ). Again, these differences between L and M2 stages (Figure 3) were unexpected. We investigated whether these differences could be due to differences in sex, using a subset of individuals for which we had this information (Table 1). Using PCA, we identified that larvae do seem



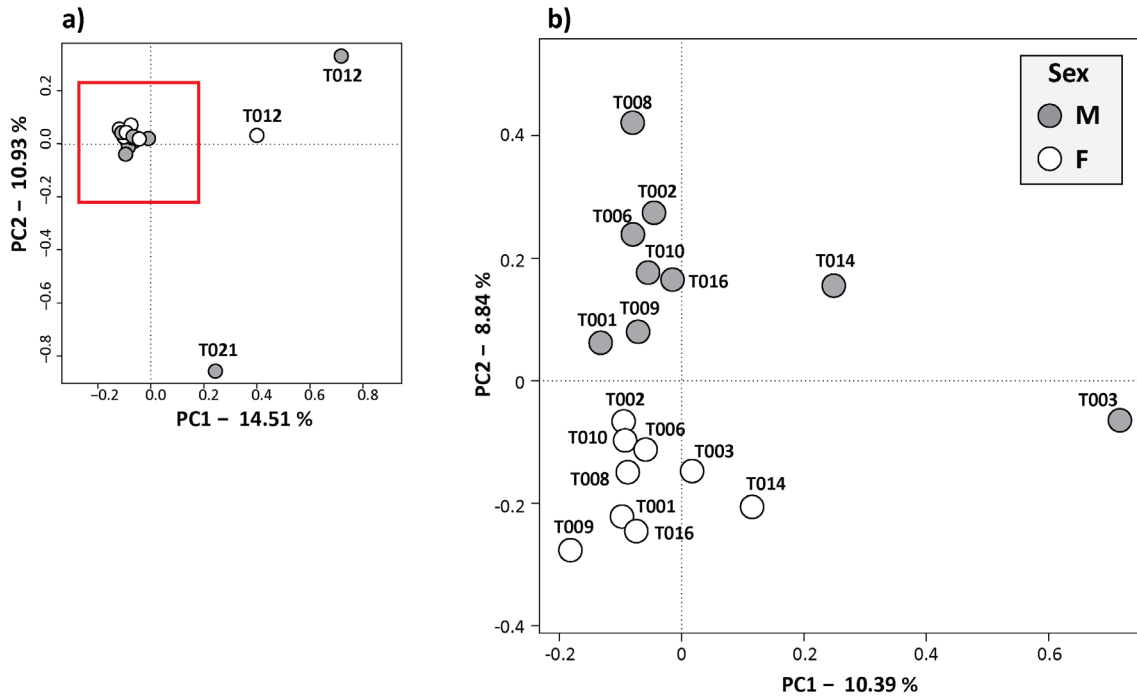
**FIGURE 3** | Principal components analysis (PCA) plot of all 2012 sampled SBW grouped by life stage. PCA was calculated using a Hellinger-transformed matrix of allele frequencies summarised to the level of each *Site-Stage* combination. Visual clustering among life stages was supported by a permutational ANOVA performed on a square root transformed pairwise matrix of *Fst* values ( $F_3 = 19.88$ ,  $p = 0.001$ ,  $R^2 = 0.59$ ). Sites with labels had fewer than five individuals. Both PC axes were significant based on the broken-stick criterion.

to differ by sex, although the ordination axes were not significant (Figure 4). Visual clustering between sexes (Figure 4) was supported by a permutational ANOVA on a square root transformed matrix of pairwise *Fst* values ( $F_{1,16} = 31$ ,  $p = 0.001$ ,  $R^2 = 0.66$ ). Differentiation between sexes was weak but significant when considering a global *Fst* using *Sex* to define populations ( $Fst = 0.005$ ,  $p = 0.01$ ). However, we did not find evidence for sex-specific clusters using DAPC (i.e., BIC was minimised at  $k = 1$ ; Figure S2).

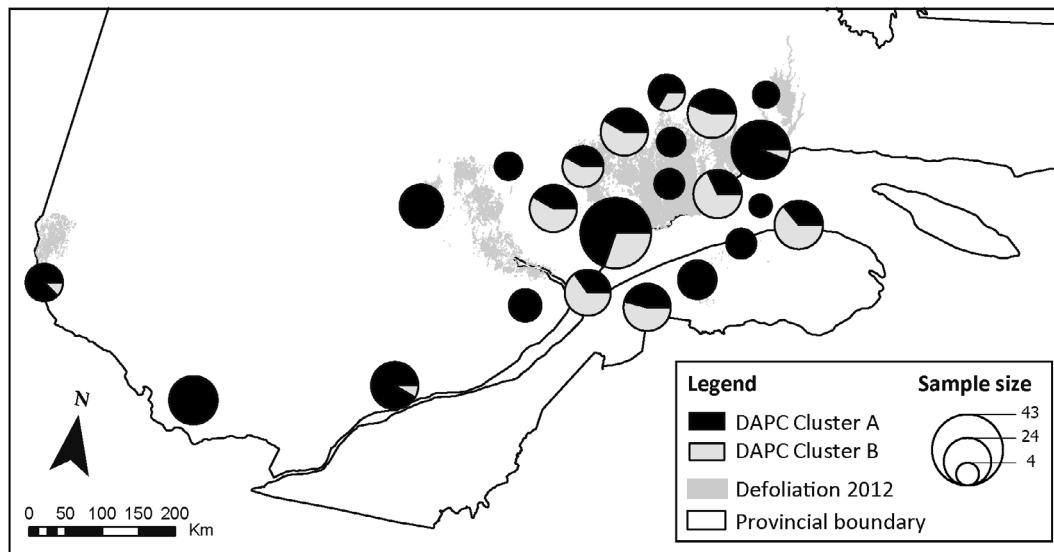
Finally, we assessed the correlation between only male larvae and M2 moths at the sites that had both groupings ( $n = 11$ ; Table 1) using a Procrustes rotation test. If the lack of concordance between resident larvae (L) and moths (M2) was due to differences in sex, we would expect the male larvae and exclusively male M2 moths to be correlated. However, again, we did not find a significant correlation ( $r = 0.55$ ,  $p = 0.154$ ) between the two groupings. Site T008 (Figure 1) seemed to have an unusually high influence on this relationship (i.e., relative highest standardised Procrustes residual). Removing site T008 from the analysis did not improve our correlation estimate ( $r = 0.53$ ), although it was closer to being significant ( $p = 0.051$ ).

### 3.3 | Clustering

When examining all individuals using DAPC, BIC was lowest for models with  $k = 2$  and  $k = 3$  (Figure S1). Because the difference in models was minimal ( $\Delta BIC < 1$ ), and  $k = 2$  was identified



**FIGURE 4** | Principal components analysis (PCA) plot of the subset of sites containing larvae for which sex-specific information was also available (Table 1). Panel (a) shows the ordination of all samples, and panel (b) shows a reduced ordination of sites found in the red square of panel (a) that exclude the highly differentiated and labelled sites. Neither axis in either ordination was significant based on the broken-stick criterion. Differentiation between male and female larvae as shown in panel (b) was supported by a permutational ANOVA performed on a square root transformed pairwise matrix of *Fst* values ( $F_1 = 31$ ,  $p = 0.001$ ,  $R^2 = 0.66$ ). Despite this, we did not find evidence of two clusters in these data using DAPC ( $k = 1$ ).



**FIGURE 5** | Spatial distribution of different SBW clusters and results from individual assignment. Plotted pie-charts illustrate the relative frequency of the two larval clusters identified using DAPC across the full network of samples sites. These two clusters were used to train a linear discriminant analysis model (*lda*) in *assignPOP* with which we assigned putative migrant and local moths to its most likely source cluster, as shown in Figure 6.

in previous studies (Larroque et al. 2019), we chose to proceed using  $k=2$  (i.e., clusters A and B).  $F_{st}$  between these two clusters was moderate and significant ( $F_{st}=0.03$ ,  $p<0.01$ ). When examining larvae only to determine the genetic structure of resident SBW,  $k=2$  was also selected. Here, genetic structure was low and significant ( $F_{st}=0.007$ ,  $p<0.01$ ). **Cluster A** individuals were found throughout the study area, whereas **Cluster B** individuals were more common in the Côte Nord region in the northeast (Figure 5).

We did not find evidence that larvae sex influences these clusters ( $k=1$  within the subset of larvae for which we had information on sex; Table 1; Figure 4). Larval clusters A and B were therefore used to assign moths to source locations. Because of the differences we identified between L and M2 grouping, we examined the number of clusters within the M2 category and found that BIC was also minimised at  $k=1$ . Therefore, we did not assign migrant moths to the M2 class—they would all be assigned to the same cluster. Similarly, BIC was minimised at  $k=1$  for M1 and M3 groupings.

### 3.4 | Population Assignment

The accuracy of assignment of individuals to source populations using *assignPOP* was consistently very high (88%–100%) and varied somewhat depending on the proportion of individuals (0.3–1.0) and the proportion of loci (0.5–1.0) used as training data. We used 100% of larval individuals and 100% of loci to train our model for assigning moths. Assignment of individuals to source clusters was dependent on life stage (Figure 4). M1 moths were only assigned to **Cluster B** (Figure 6a), M2 moths were assigned more often to **Cluster B** (Figure 6b), and M3 moths were split between **Clusters A** and **B** (Figure 6c). Using a Chi-square test, we confirmed that there was a significant difference in assignment probability among the three moth categories, which was most strongly

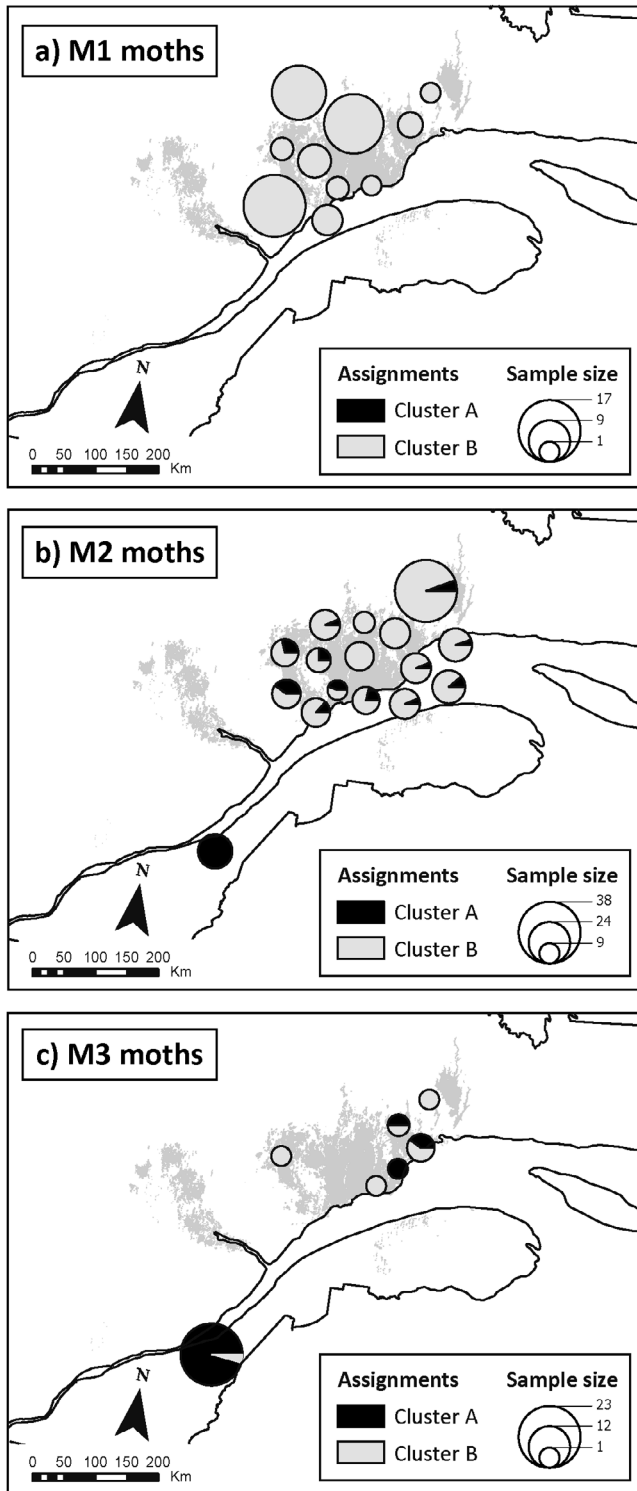
driven by differences in the M3 class (Table 4;  $\chi^2=84.04$ ,  $df=2$ ,  $p<0.001$ ). Using Fisher's exact test, we also found that the assignment of M2 moths did not correspond to the larval clusters. Larvae were more likely to be assigned to **Cluster A**, whereas M2 moths were more likely to be assigned to **Cluster B** (Odds-Ratio = 10.18,  $p<0.001$ ).

Because the M2 and M3 groupings at the Armagh site were not well differentiated (Figures 1 and 3), we considered the possibility that they were all from the same local (M2) moth group. We evaluated the sensitivity of our interpretations to the removal of this site and found that without Armagh, for which we had far more migrant moths than for any other site, there was only a modest difference in the probability of being assigned to **Cluster A** ( $n=4$ ) or **B** ( $n=7$ ). However, using the Chi-square test, we still found significant differential cluster assignment among moth categories ( $\chi^2=16.86$ ,  $df=2$ ,  $p<0.001$ ).

## 4 | Discussion

The goal of this study was to better characterise patterns of dispersal and connectivity in spruce budworm during an outbreak. We examined genetic diversity and differentiation among larvae and adult moth spruce budworm at three different timings of collections. When considering all data, only observed heterozygosity exhibited a significant difference between groupings. When we removed the M3 class due to the limited number of sites at which this category was collected ( $n=2$ ), we identified significant differences among groupings in all three genetic metrics. These differences were consistently driven by differences between larval (L) and resident moths (M2) groupings.

We also identified weak geographic genetic structure for all groupings when considering sample locations (i.e., sites;



**FIGURE 6** | Spatial distribution of moth assignment using *lda* model developed using the two DAPC-identified clusters illustrated in Figure 5. Panels a-c illustrate the proportion of moths assigned to each of the two clusters at each sample location. M1, M2 and M3 refer to early migrant moths, resident moths and late migrant moths, respectively.

Table 1) as putative populations. Previous work has identified weak structure and high connectivity among larvae (Larroque et al. 2019), but this is the first study to report on connectivity based on adult moths, which is much more complex given their greater capacity for dispersal. Together, these results indicate a

**TABLE 4** | Individual Chi-square values and number of individuals assigned to each category (in parentheses). Migrants were assigned between clusters in proportions that differed from expectation.

| Stage                                            | Clusters   |            |
|--------------------------------------------------|------------|------------|
|                                                  | A          | B          |
| M1                                               | 12.98 (0)  | 3.12 (67)  |
| M2                                               | 1.02 (46)  | 0.24 (235) |
| M3                                               | 63.27 (25) | 15.20 (9)  |
| Total $\chi^2 = 95.827$ , $df = 2$ , $p < 0.001$ |            |            |

high degree of connectivity among populations in both larvae and adults. This result conforms with theoretical expectations of an increasing, near-peak epidemic phase of an outbreak, where high population densities and likely high rates of density-dependent dispersal result in a high degree of gene flow (James et al. 2015).

Despite these indications of panmixia, we identified some interesting differences among SBW groupings when examining an ordination of site-level allele frequencies (Figure 3). We found significant clustering of sites based on life stage, which accounted for nearly 60% of genetic variation. Putative early and late migrants appear distinct from each other and from the L and M2 stages, although their unique locations in the PCA biplot may be affected by reduced sample sizes. Larvae were also distinct from resident M2 moths, in contrast to our expectation that resident moths and larvae represent two samples of the same local populations.

One potential explanation of this lack of concordance between L and M2 groupings is that M2 moths were not as local as our phenological model indicated. That is, M2 moths may have migrated from an area slightly out of phenological synchrony with the local site and were hence misclassified as phenologically synchronous residents. However, the absence of strong geographic differences among larvae (e.g., Larroque et al. 2019; Figure 3), even those separated by large distances, suggests that migration from nearby regions in near, but not complete, phenological synchrony is unlikely to explain genetic differences at a site.

The other hypothesis is that there are important differences in dispersal or other attributes between male and female moths. All collected moths are male, whereas larvae are both male and female, although the sex of all individuals is not known. Previous work on population genetic structure examined larvae without regard for sex (Larroque et al. 2019, 2020; James et al. 2015) and sex-specific differences in spatial genetic variation remain unexamined in this system. Sex-biased dispersal has been observed in other insect species (Petit and Excoffier 2009), such as outbreeding Coleopterans (Safranyik and Wilson 2006) and Lepidopterans (Legrand et al. 2016), including the SBW (Rhains et al. 2022). Further, female SBW are known to deposit a portion of their egg complement prior to flight events (Rhains and Kettela 2013), potentially further complicating efforts to infer population connectivity in this highly mobile, outbreeding species.

In analysing a small subset of sexed larvae, we identified significant differences between males and females. Here, sex accounted for 66% of variation among individuals. In comparing male larvae with captured male moths, we expected to see a stronger correlation. However, once again, this correlation was weak and non-significant. These results do not help to clarify the lack of correlation between resident larvae and putatively resident moths, and it remains uncertain why these two groupings were not more similar. However, these results do indicate that interesting differences between male and female spruce budworm exist in terms of dispersal or genetic structure, which should be investigated further.

#### 4.1 | Assignment

Although there was weak structure associated with sample locations, we identified two distinct resident larval clusters (Figure 5). This  $k = 2$  solution in larvae was previously described by Larroque et al. (2019). We expanded on this previous work and investigated the cluster from which moths collected at different times within a season had likely originated. Larval **Cluster A** was widespread across the study area, whereas **Cluster B** appears more common in the east, with some representation in the western part of the study area.

Early migrant moths were all assigned to **Cluster B**, whereas late migrants showed a mixture of the two clusters. Resident moths were also mixed between clusters, but more individuals were assigned to **Cluster B**, which was interesting given that larvae were more often assigned to **Cluster A**. We found strong evidence that the assignment of moths to clusters was different from what would be expected by random chance (Table 4). These differences were most strongly driven by the absence of any early migrants assigned to **Cluster A** and more than expected late migrants assigned to **Cluster B** (Table 4). It is unknown if moths from **Clusters A** and **B** are functionally different in any way.

The lack of cluster fidelity to sampling locations, or even regions, was surprising. Overlap of the two clusters suggests the previous existence of two distinct populations that merged. These two populations could hypothetically represent the areas to which the previous outbreak collapsed and that had the first increases in population growth at the beginning of the outbreak. The outbreak was first detected (via defoliation survey) in the Côte Nord region of Quebec (Bouchard and Auger 2014) where **Cluster A** seems to be more common. We speculate that if we had sampled several years earlier, prior to population homogenisation via dispersal, geographic association with specific clusters may have been more apparent. Large-scale mass dispersal events are known to occur in spruce budworm populations at high densities (Boulanger et al. 2017; Greenbank et al. 1980) which can homogenise genetic variation, thus masking the genetic legacies (i.e., bottlenecks) of the previous outbreak (James et al. 2015). These results and associated interpretations are consistent with the ‘epicentre’ hypothesis of outbreak synchrony. Ongoing population and genetic monitoring as the outbreak subsides holds promise to better elucidate how outbreaking population dynamics influence spatial and temporal genetic variation.

Results of this study are important for our theoretical understanding of SBW outbreaks but also have implications for how SBW is managed. To mitigate SBW-induced damage, forest management has traditionally relied on reactive measures such as the aerial spraying of pesticides (Hennigar et al. 2007; Kettela 1975). More recently, efforts have shifted towards a proactive control strategy (MacLean 2019) that targets low-density areas that have the potential to develop into outbreak epicentres (Johns et al. 2019). Not only has this Early Intervention Strategy (EIS) been proven successful in mitigating outbreak spread in New Brunswick (MacLean 2019), EIS has a higher benefit–cost ratio than traditional foliage protection strategies (Liu et al. 2019). Because EIS is most effective in populations that have yet to reach outbreak levels, it is essential to understand local population dynamics including how frequently individuals disperse and from how far. If dispersal is not relevant to synchronous outbreaks, then EIS is unlikely to be successful. In contrast, if there is evidence for a strong role of dispersal in outbreak spread, then one can expect EIS to have a great probability of success, depending on the interannual frequency of moth immigration, successful reproduction at arrival locations, and our ability to rapidly and meaningfully detect hotspots of rapidly increasing population density. Although we were not able to assign migrants to specific source locations and evaluate dispersal parameters (e.g., distance), we were able to demonstrate that there are complex patterns of movement during an outbreak peak within a season. Future studies that examine genetic variation and population connectivity during the declining and endemic outbreak phases will help to further address these outstanding questions. Further, given the long time scales and large spatial extent over which SBW outbreaks occur, we recommend the development and application of novel demographic simulation models (e.g., Mayrand et al. 2019) to help generate expectations and test hypotheses regarding the effects of cyclic outbreak dynamics on genetic variation and inference.

#### 4.2 | Study Limitations

We used pheromone traps and information on budworm phenology to distinguish resident from migrant moths. However, there is some uncertainty in the final classifications due to uncertainty in the phenological model and timing of moth collection. Challenges can arise when using BioSIM to interpolate far from weather stations (Dirks et al. 1998; Mair and Fares 2011; Ly et al. 2013). Consequently, some moths may not have been classified correctly. Future work will include examining the accuracy and spatial scale of uncertainty in these phenological predictions in remote areas such as the boreal forest, assessing the sensitivity of our assignment model to such variation, and investigating alternate methods for weather interpolation. Also, there are opportunities to integrate these genetic assignment methods with other spatial demographic models (Gandiaga and James 2023).

Another potential limitation is the timing of moth collection. Moths were collected from pheromone traps three times per location through the flight season over a very large spatial extent. Three times was the maximum possible for a single field

crew and involved 10s of thousands of kilometres of driving in remote areas. Traps were emptied up to 2 weeks after being initially installed or reinstalled after emptying. As a result, the date on which samples were collected from the trap may not reflect the exact date of trap entry. The delay between arrival and sampling combined with the final observed phenology could have resulted in some residents being classified as migrants, and vice versa. However, given our experimental design and the coarse timing of initial collection we used, followed by our careful final assignment based on observed phenology and known dates of trap installation, we do not expect that many moths were misclassified. Further, the differences we successfully identified among groupings (e.g., Figure 3) lend support to the accuracy of our final classifications. Future work in this area will refine estimates of trap arrival timing using daily trap captures from the citizen science Budworm Tracker programme (Carleton et al. 2020).

## 5 | Conclusions

Outbreaks are fascinating and complex phenomena driven by multiple processes at different scales. The application of molecular markers continues to help us better understand the inter- and intra-annual processes that drive synchronous outbreaks. We demonstrated a novel approach to distinguishing resident from migrant moths and assigning migrants to most likely source locations using genotypic information. We demonstrated a high degree of population connectivity across a huge area of ongoing outbreak and that migrant moths do not all originate from the same clusters. Improved understanding of the role of dispersal in outbreak dynamics is important to support and inform management strategies that aim to control incipient hotspots of high population density before the outbreak becomes too extensive. Finally, this work also highlights a significant gap in our understanding of how cyclic irruptive dynamics affect population genetic inference. Further research in this area holds potential to improve management of outbreaking or invading populations.

### Author Contributions

P.M.A.J. conceptualised, designed and performed the research, secured funding for the project, analysed the data and wrote the paper. J.T.F. performed research, helped analyse data and helped write the paper. J.W. performed research, contributed analytical tools and helped write the paper. M.C. contributed data and helped write the paper. J.L. contributed analytical tools, processed the data and helped write the paper. A.R. helped design the research and helped write the paper. R.J. helped design the research and helped write the paper.

### Acknowledgements

This study was supported by an NSERC Discovery Grant to P.M.A.J. and the ACOA Early Intervention for Spruce Budworm Outbreaks program. We thank the numerous field and laboratory technicians who helped collect samples and generate the genomic data.

### Disclosure

Benefit-Sharing Statement: Benefits from this research accrue from the sharing of our data and results on public databases as described above.

### Conflicts of Interest

The authors declare no conflicts of interest.

### Data Availability Statement

Raw sequence reads are available from the NCBI Sequence Read Archive (SRA) under accession number: PRJNA1265759; <https://www.ncbi.nlm.nih.gov/bioproject/1265759>. SNP files, metadata and scripts for all analyses are available on Zenodo at doi: [10.5281/zenodo.15467607](https://doi.org/10.5281/zenodo.15467607).

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## Supporting Information

Additional supporting information can be found online in the Supporting Information section.